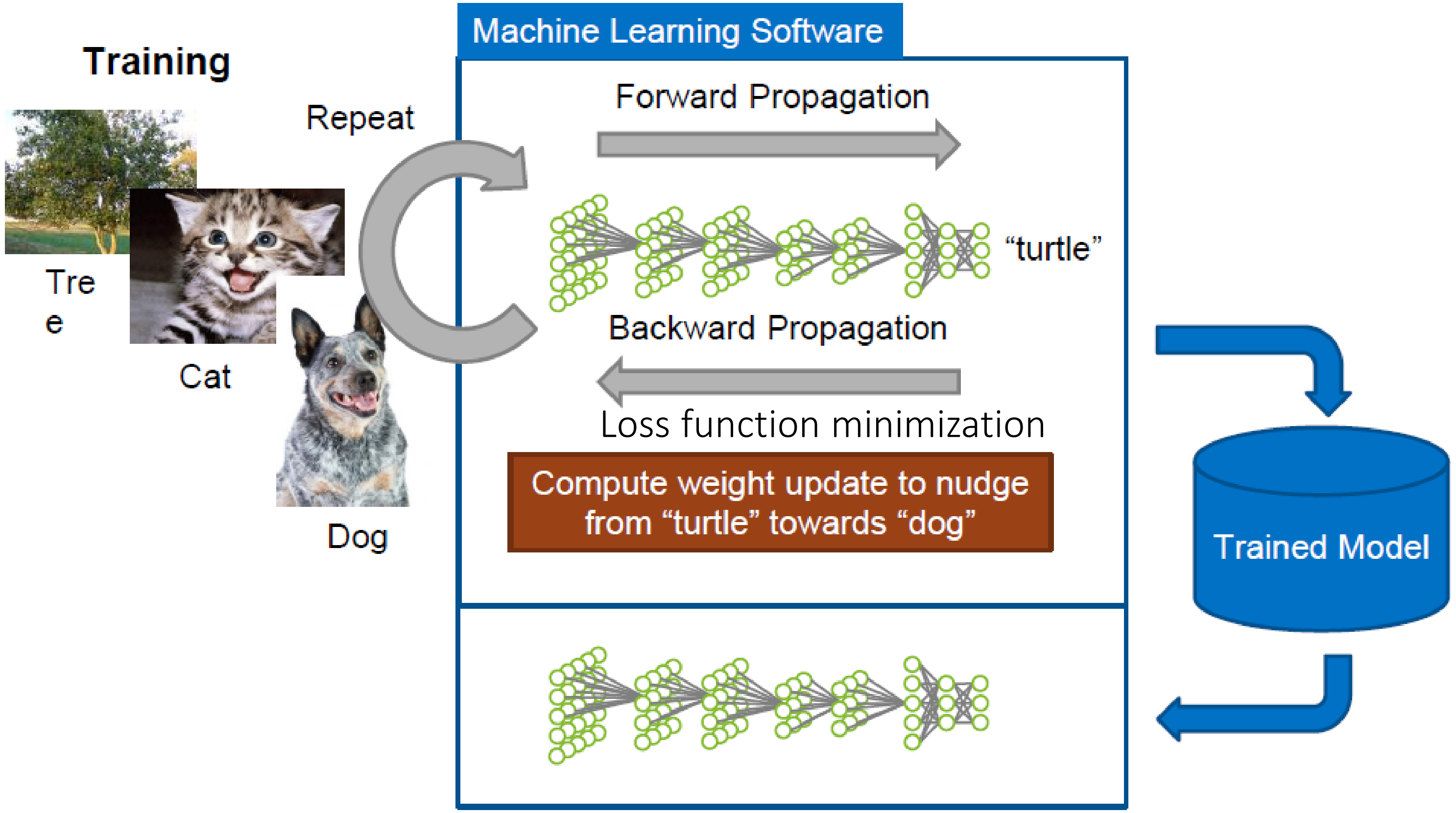
Biomedical Image Processing

DEEP LEARNING AND SEGMENTATION

Helena R. Torres

Deep Learning



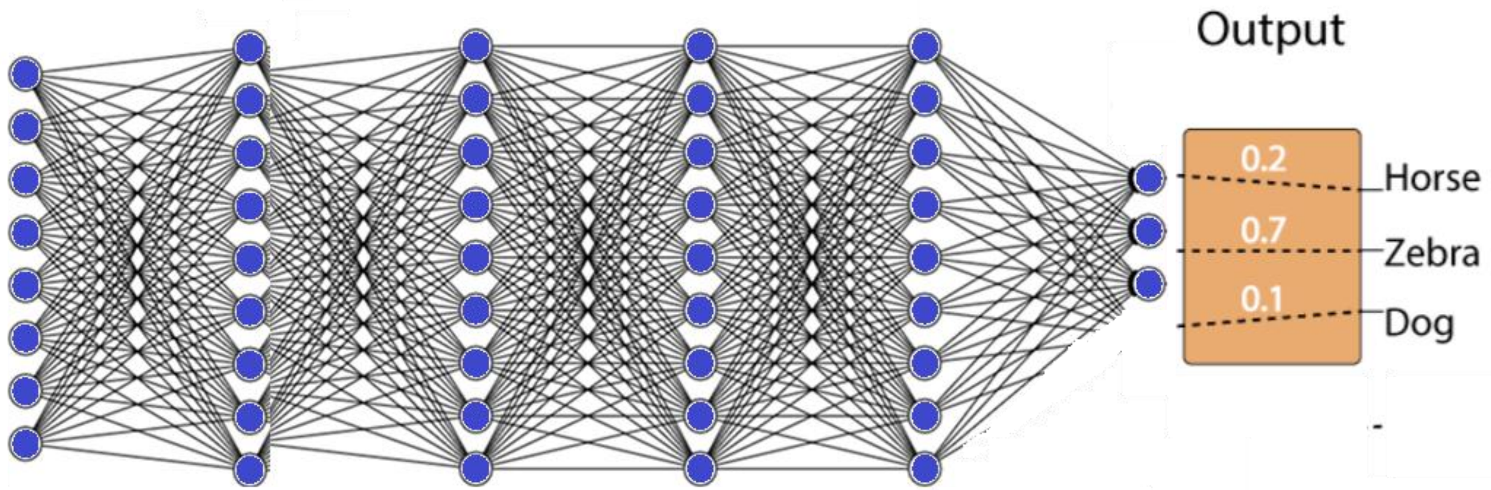
LAYERS

TRADITIONAL NEURAL NETWORK VS CONVOLUTIONAL NETWORK

Fully connected networks

- Each neuron in one layer is connected to every neuron in the next layer;
- Have a very high number of parameters;
- Computational expensive.



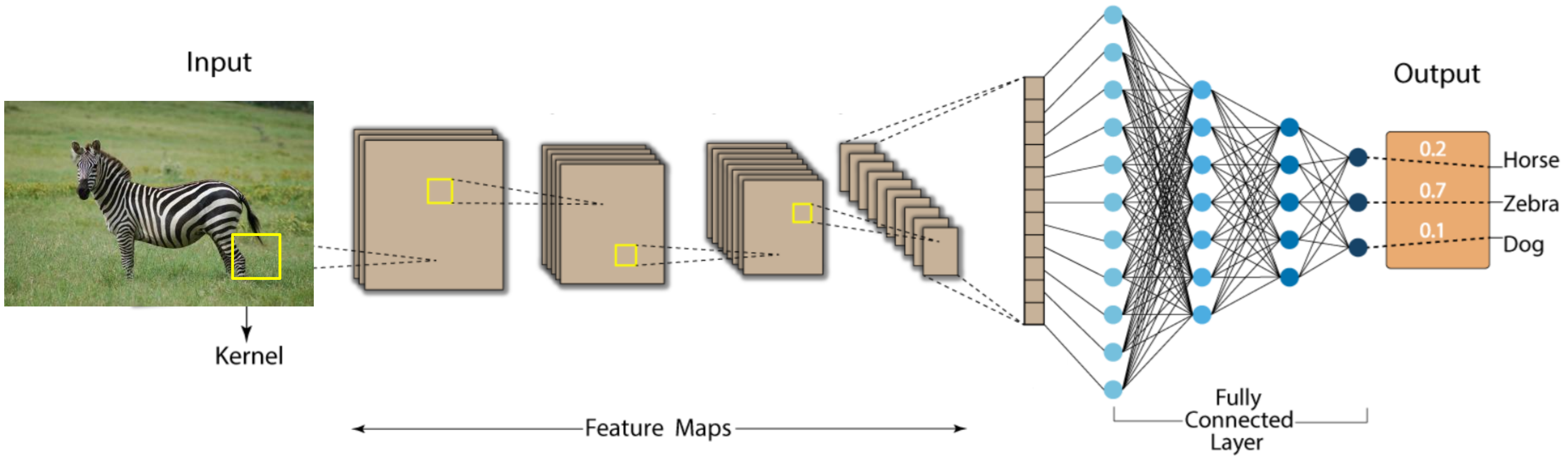


LAYERS

TRADITIONAL NEURAL NETWORK VS CONVOLUTIONAL NETWORK

Convolutional networks

- Use convolutional layers with filters (kernels) that slide across the input;
- Use shared weights (filters), drastically reducing the number of parameters;
- Preserve spatial relationships.

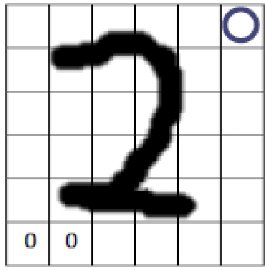


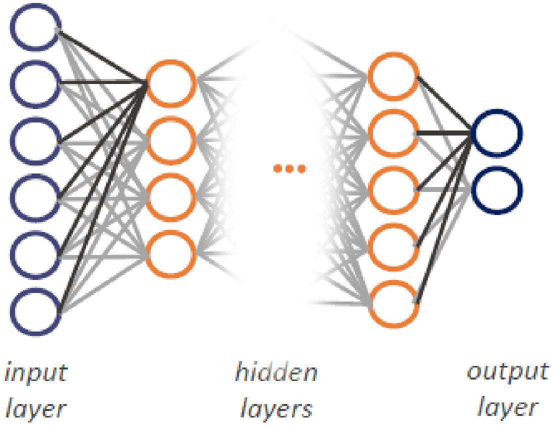
LAYERS

CONVOLUTION LAYERS vs NEURAL NETWORKS

Neural networks:

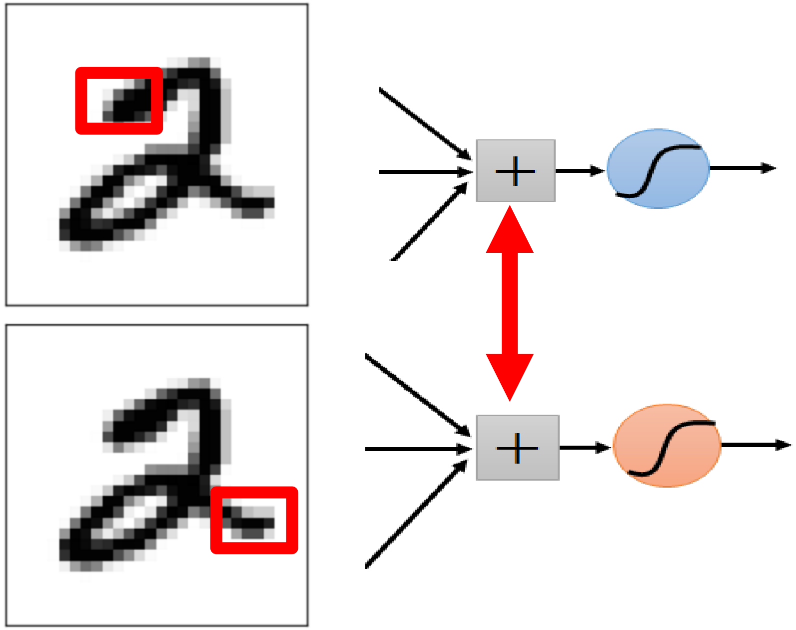
- Analyzes the image as a whole;
- Puts all the pixels in a long list and passes them through the network simultaneously.





Convolutional networks:

- It focuses on small parts of the image at a time;
- Leads the network to use the information in a specific way.



LAYERS

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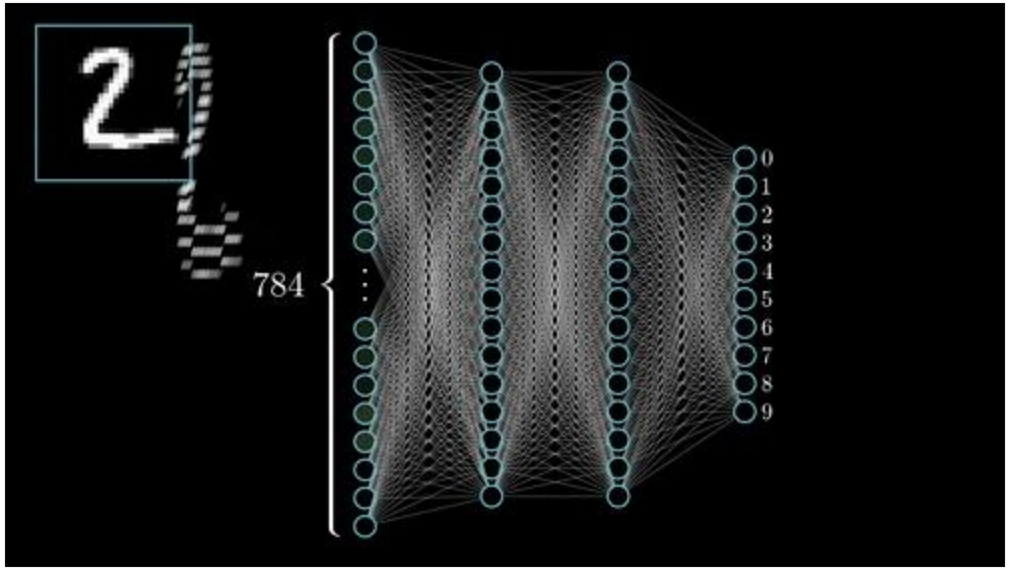

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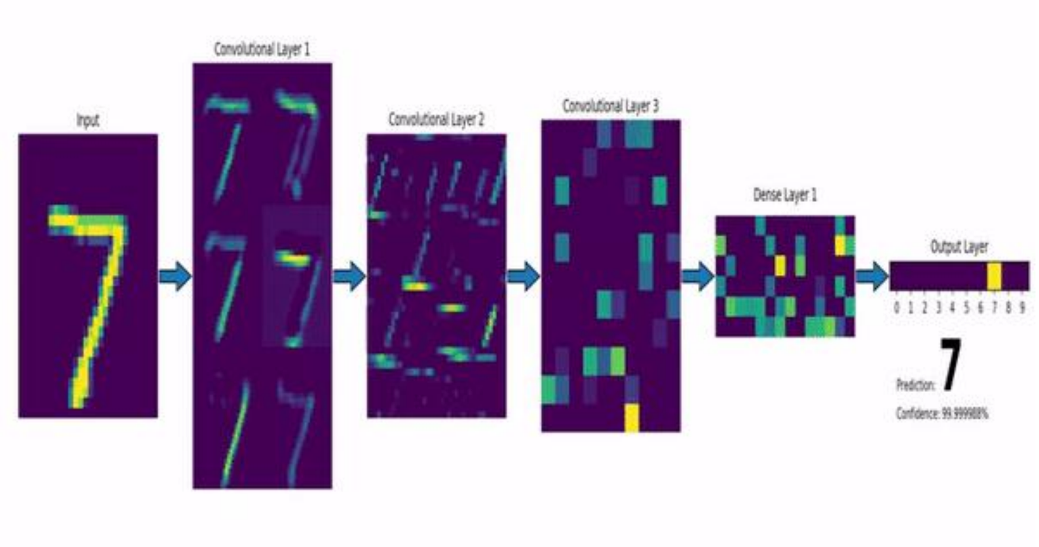
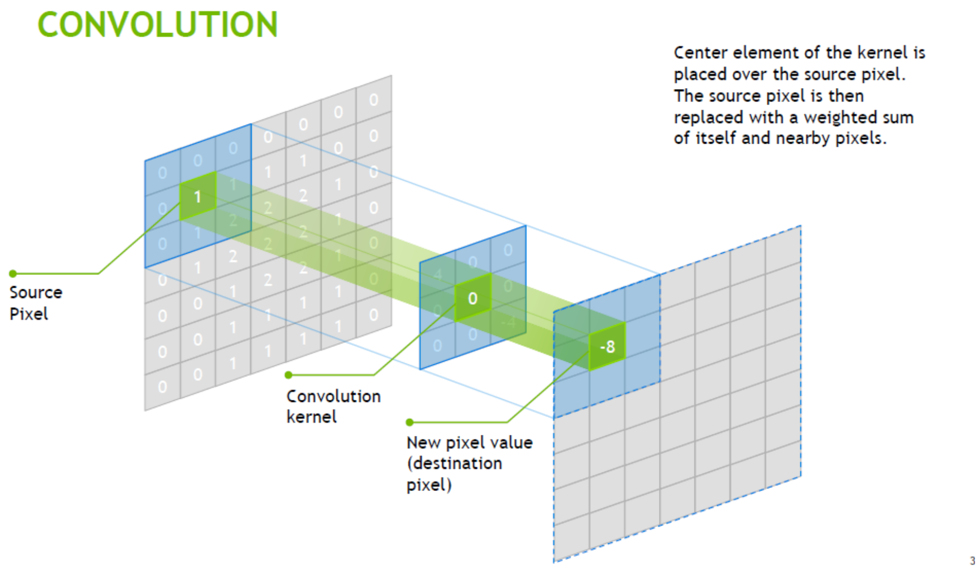
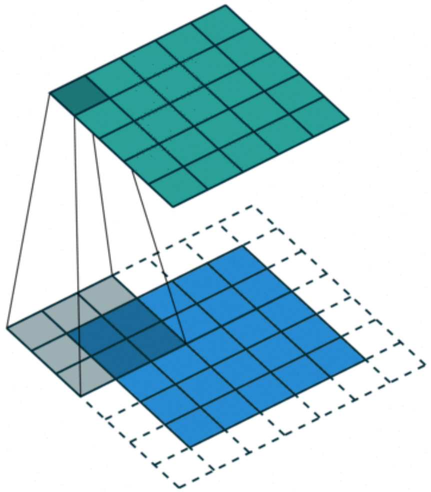


LAYERS

CONVOLUTION LAYERS

- A convolution layer presents a set of filters that perform a convolution operation on the entire image in order to activate image-specific features;
- They work as if they were "little detectors" that run through the image.





Originate feature maps!

LAYERS

CONVOLUTION LAYERS

❑ *Stride*:

- Corresponds to the number of pixels that the filter shifts;
- The smaller the stride, the greater the overlap in the generated feature map;
- The higher the stride, the less overlap there will be on the feature map and the smaller this map will be.

❑ *Padding:*

- Corresponds to the zeros placed around the image so as not to lose the features at the edge of the image.

LAYERS

CONVOLUTION LAYERS - EXAMPLE

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

Image 6 x 6

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

-1	1	-1
-1	1	-1
-1	1	-1

Filter 2

⋮ ⋮

Each filter detects a pattern (3 x 3)

Each parameter of the filters is what will be learned by the network!

LAYERS

CONVOLUTION LAYERS - EXAMPLE

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

stride=1

3	-1	-3	-1
-3	1	0	-3
-3	-3	0	1
3	-2	-2	-1

Features Map

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

LAYERS

CONVOLUTION LAYERS - EXAMPLE

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

Repeat for each filter

-1	-1	-1	-1
-1	-1	-2	1
-1	-1	-2	1
-1	0	-4	3

-1	1	-1
-1	1	-1
-1	1	-1

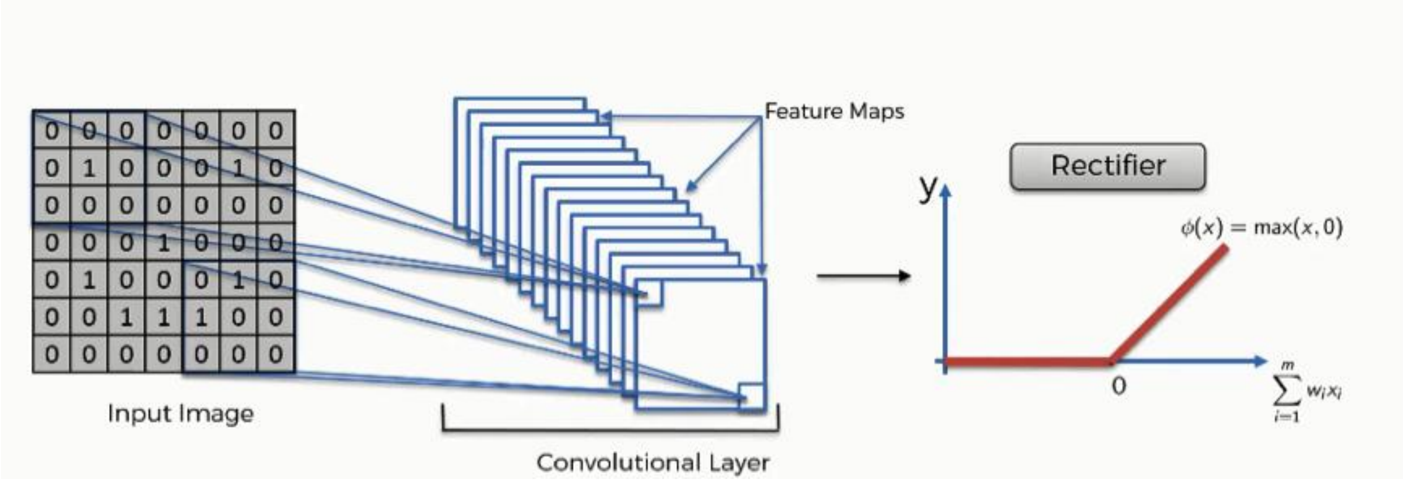
Filter 2

Results in two images 4x4

LAYERS

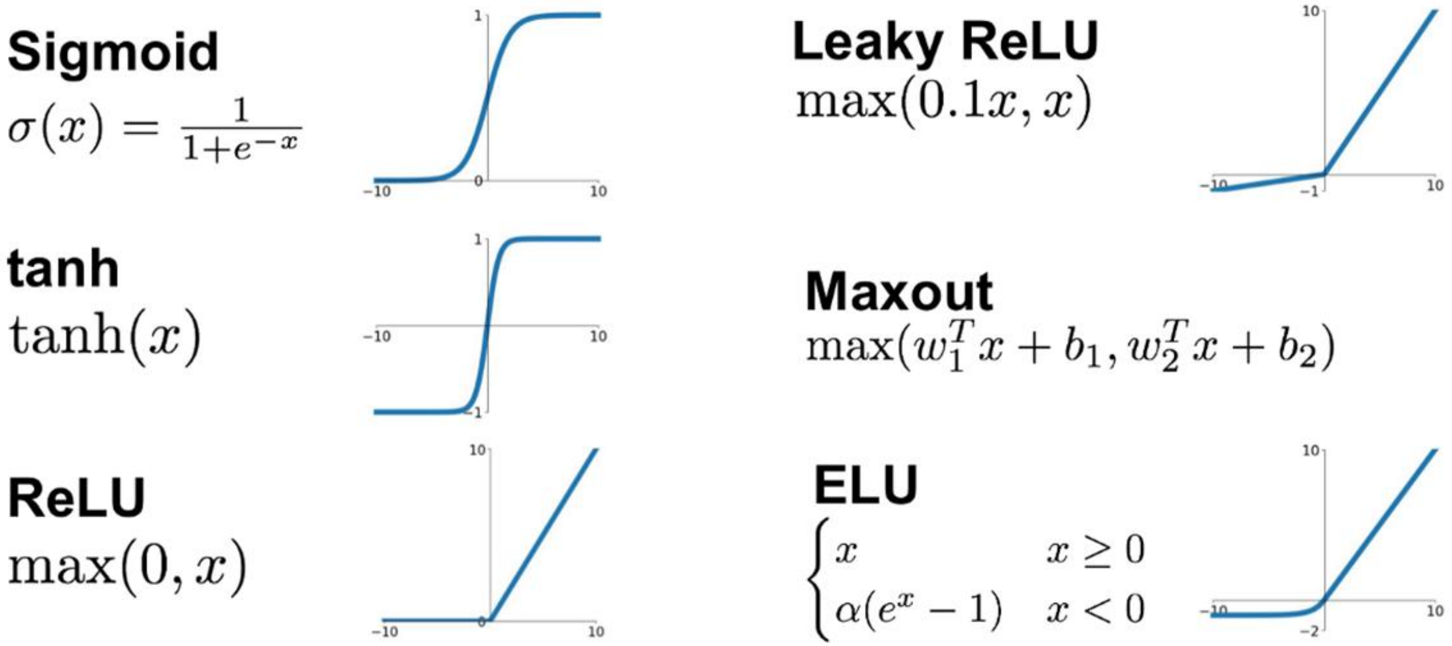
LAYERS RELU (*Rectified Linear Units*)

- After each convolution layer, it is common to apply a nonlinear layer (or activation layer);
- The purpose of this layer is to introduce non-linearity to the system that basically so far has only performed multiplications and sums;
- Negative activations are transformed into zeros.



LAYERS

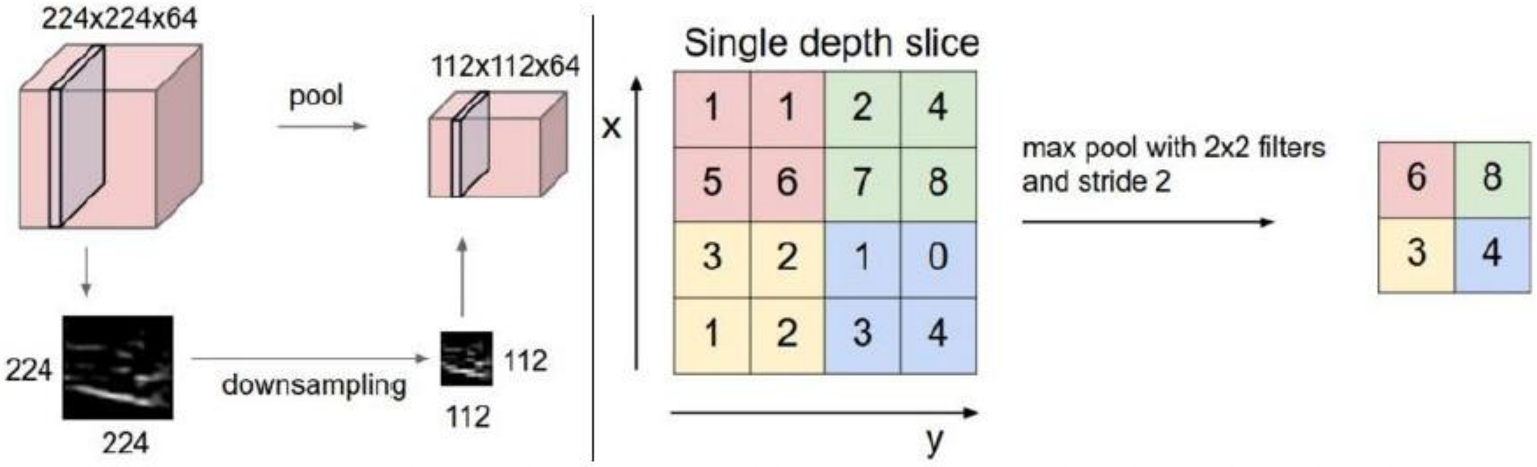
OTHER ACTIVATION LAYERS

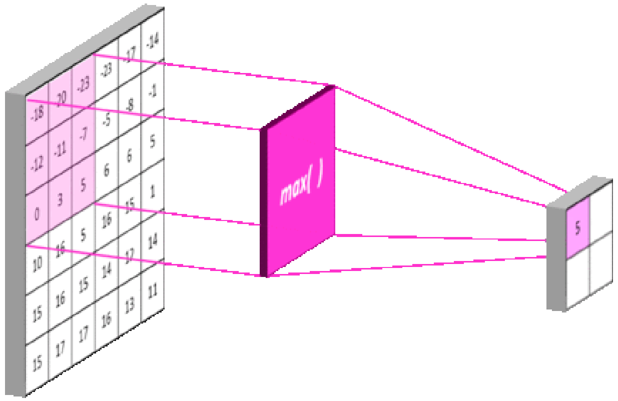


LAYERS

POOLING LAYERS

- Layer that applies downsampling to feature maps generated by previous layers;
- They may carry out maximum, average or normalisation operations;
- Allows you to reduce the computational cost of the network.

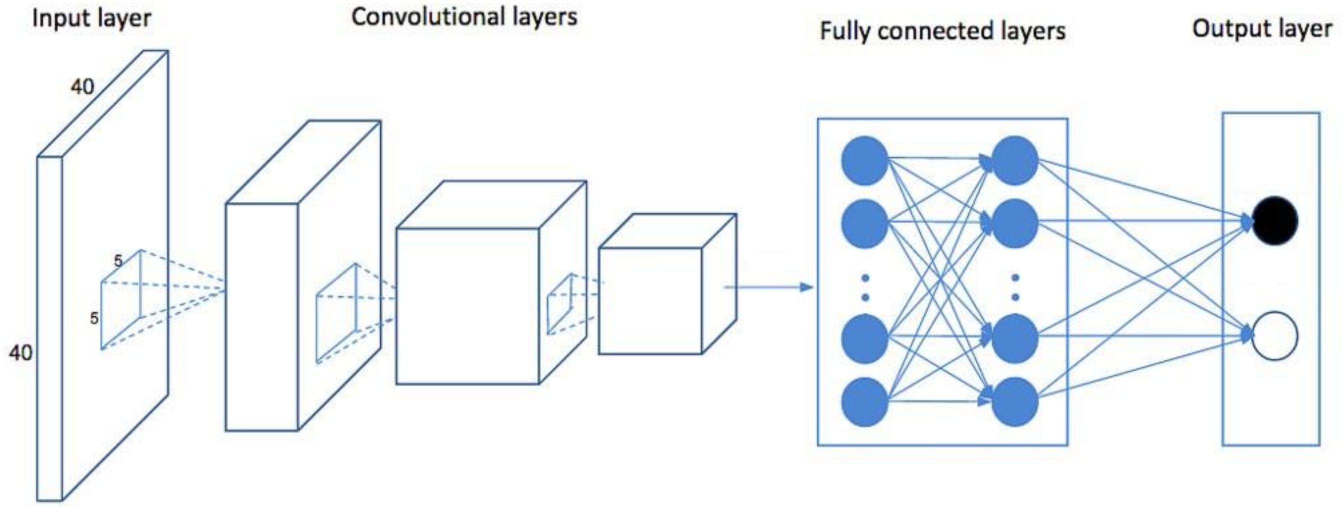




TIPOS DE *LAYERS*

LAYERS FULLY CONNECTED

- Transforms the output of a convolution layer or RELU into an N-dimensional vector where N corresponds to the number of classes through which the program must classify the image;
- Analyzes the output of a layer and determines which features are characteristic of each class;
- In this way, we obtain the probability that the feature maps correspond to a given class.

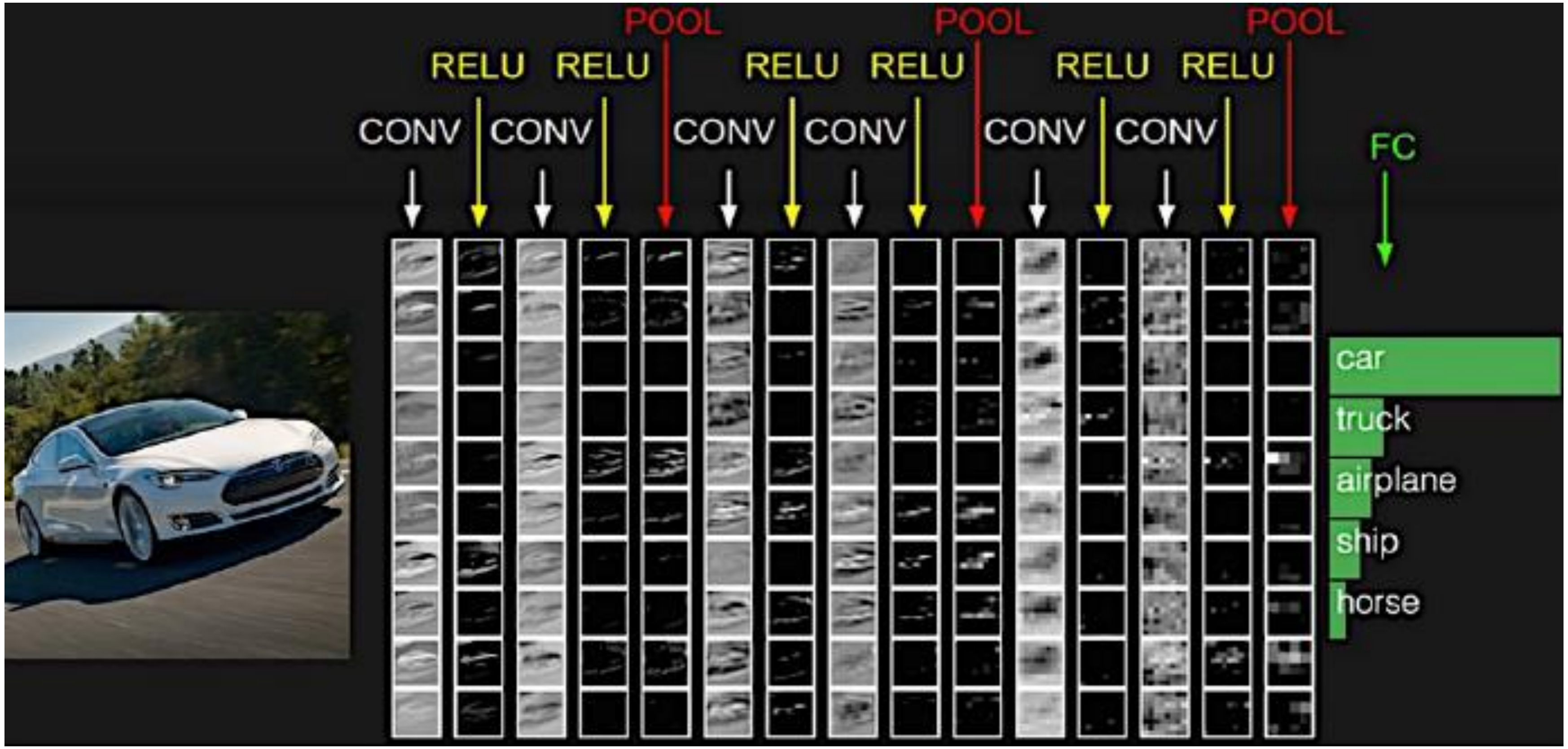


Deep Learning

EXEMPLO

Classification

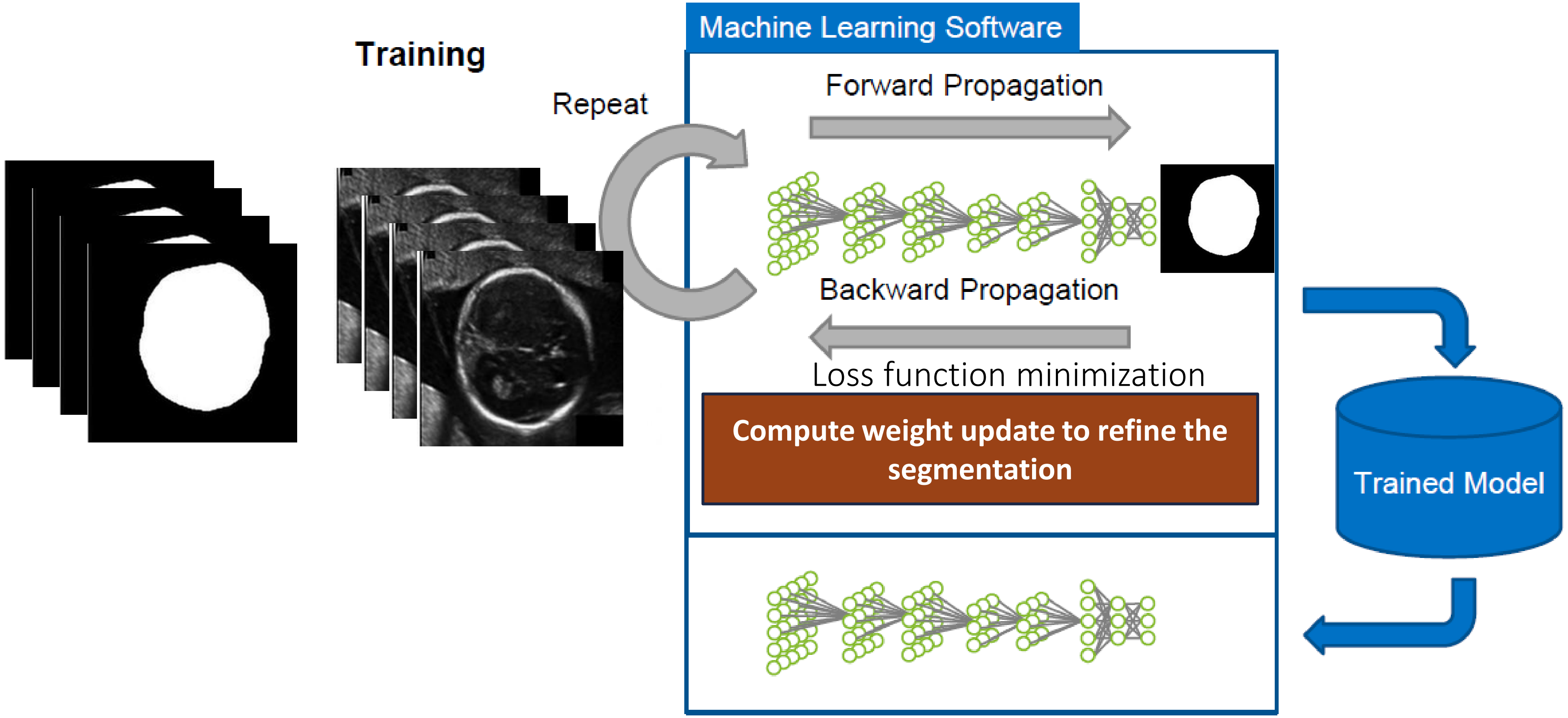
https://poloclub.github.io/cnn-explainer/



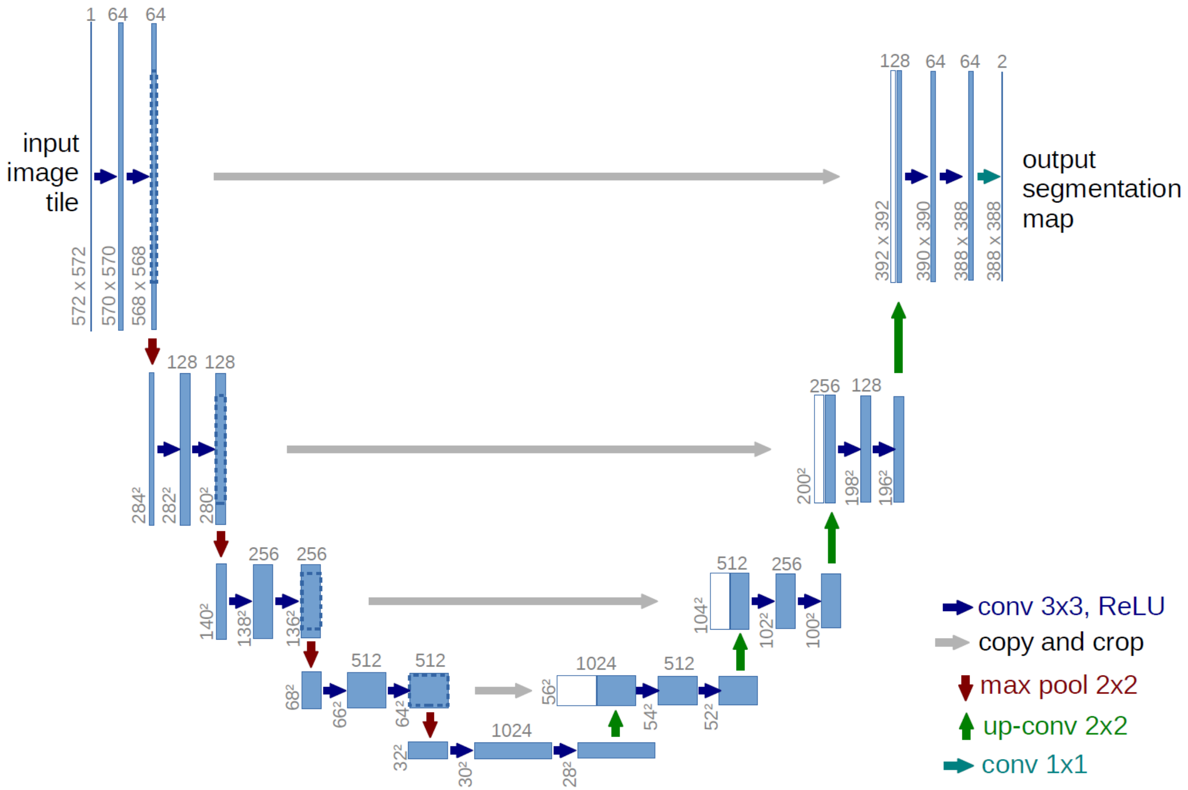
Standard networks

- Classification
  - AlexNet, VGG, GoogleNet, ResNet, MobileNet
- Object detection
  - R-CNN, Fast RCNN, Faster RCNN, Mask RCNN, YOLO, SSD
- Segmentation
  - UNet, Segnet, DeepLab, FCN, DeconvNet, Mask-RCNN

Deep Learning - Segmentation



Deep Learning – Segmentation UNet

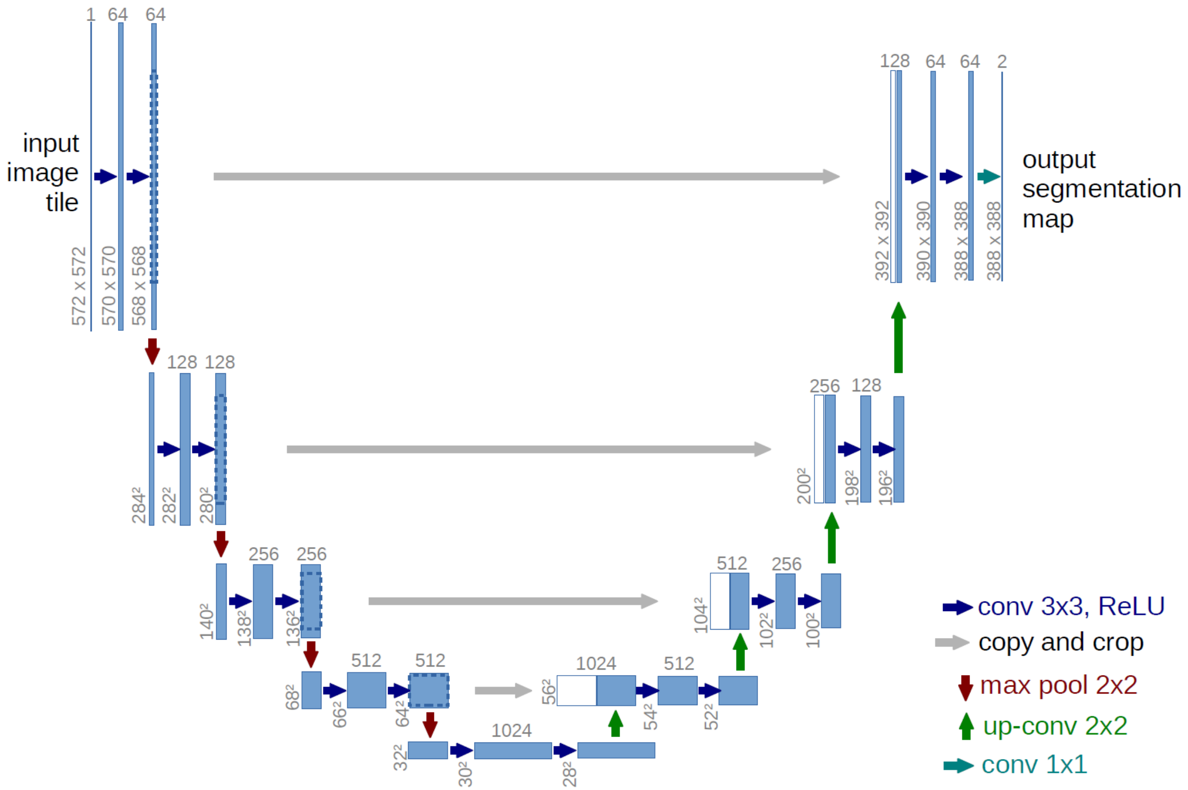


•**Encoder-decoder architecture**: Symmetrical structure with downsampling (contracting) and upsampling (expanding) paths.

•**Contracting path (encoder):** Captures context using repeated convolution and max pooling to extract features and reduce spatial dimensions.

•**Expanding path (decoder):** Reconstructs spatial resolution using up-convolutions and concatenates features from the encoder via skip connections for precise localization.

Deep Learning – Segmentation UNet



•**Encoder-decoder architecture**: Symmetrical structure with downsampling (contracting) and upsampling (expanding) paths.

•**Skip connections**: Connect corresponding layers from encoder to decoder for precise localization.

•**Convolutional layers**: Uses repeated 3x3 convolutions followed by ReLU activations.

•**Max pooling**: Downsamples feature maps in the encoder.

•**Up-convolutions**: Upsample feature maps in the decoder using transposed convolutions.

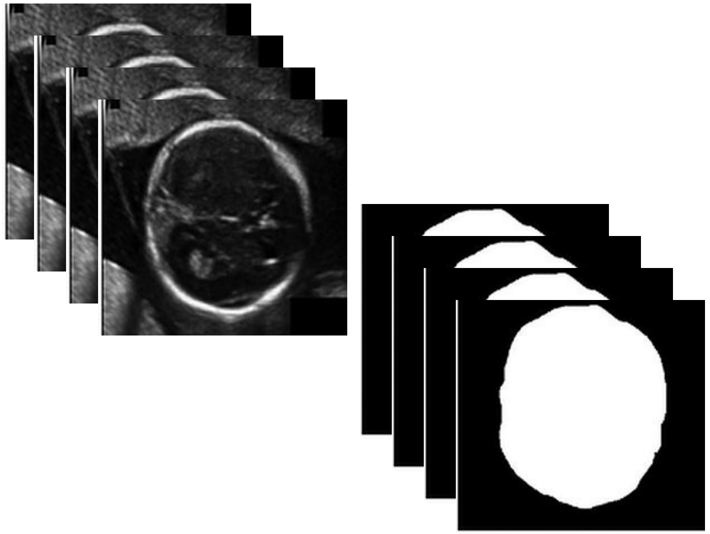
Deep Learning in Medical Imaging

MONAI PROJECT

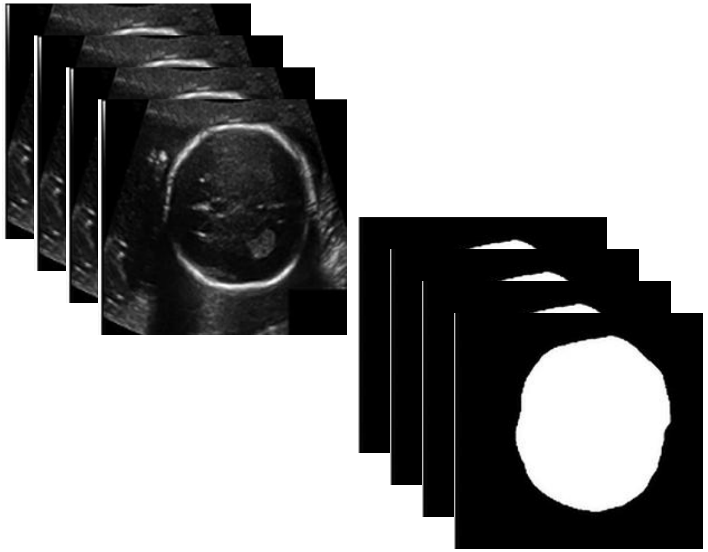
- https://monai.io/
- https://github.com/Project-MONAI
- https://github.com/Project-MONAI/tutorials
- https://colab.research.google.com/github/Project-MONAI/tutorials/blob/master/2d_classification/mednist_tutorial.ipynb#scrollTo=KQUqy2q9vq25
- https://colab.research.google.com/github/Project-MONAI/tutorials/blob/master/3d_segmentation/spleen_segmentation_3d.ipynb#scrollTo=JYtbhjtLv9qV

How to train a model for segmentation?

Training set

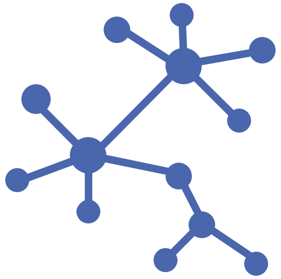


Validation set

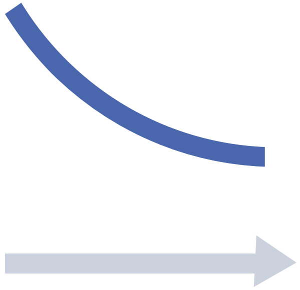


Testing set

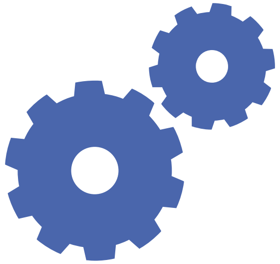
How to train a model?



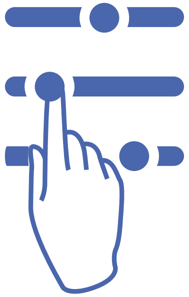
Network model



Loss function



Optimizer



Parameters
(learning rate, batch size, epochs)

How to train a model?

Traditional workflow

Dataset preparation

Split the dataset into training, validation, and testing

Define data augmentation strategies

Model definition

Develop/choose the network configuration

Define the training parameters and the loss function

Train model

Execute the training processing in the training dataset…

while validating the model in the validation dataset

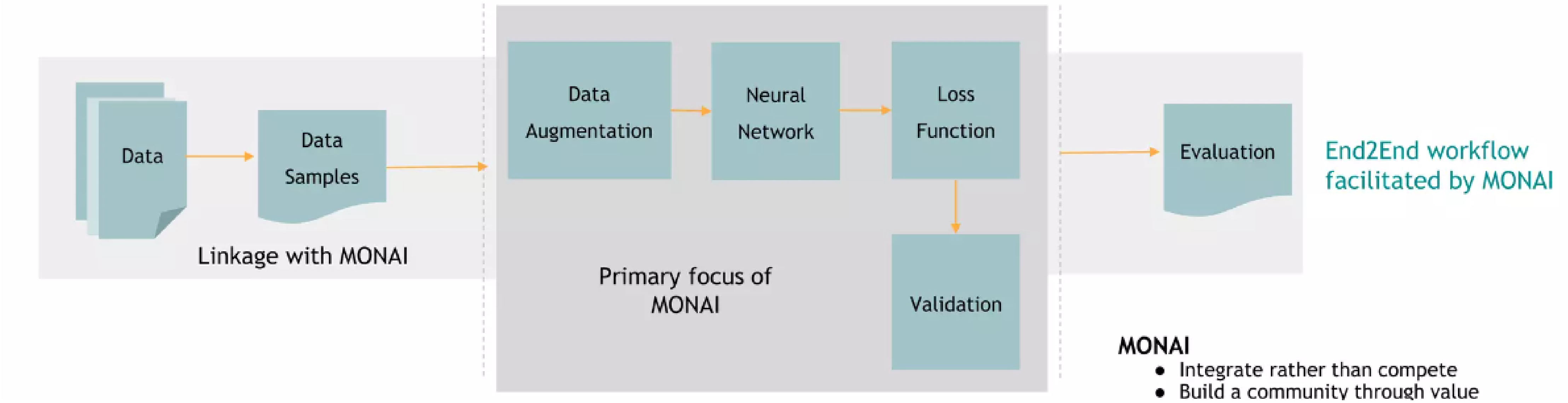
Test in new images

Once trained, apply the model to new images to generate the output

How to train a model?

Medical Open Network for Artificial Intelligence

Dataset preparation | **Model definition** | **Train model** | **Test in new images**



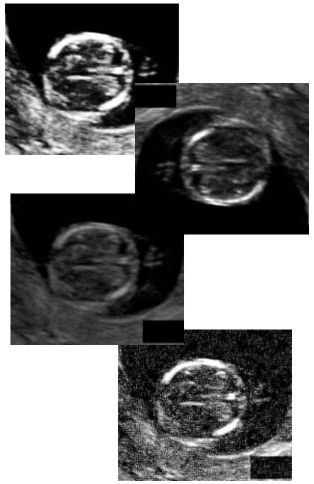
MONAI

Medical Open Network for Artificial Intelligence

How to train a model?

Dataset preparation
Load images
Transforms

Vanilla Transforms
- Crop and Pad
- Intensity
  - *RandGaussianNoise*
    - `RandGaussianNoise`
  - *ShiftIntensity*
  - *RandShiftIntensity*
  - *StdShiftIntensity*
  - *RandStdShiftIntensity*
  - *RandBiasField*
  - *ScaleIntensity*
  - *RandScaleIntensity*
  - *ScaleIntensityFixedMean*



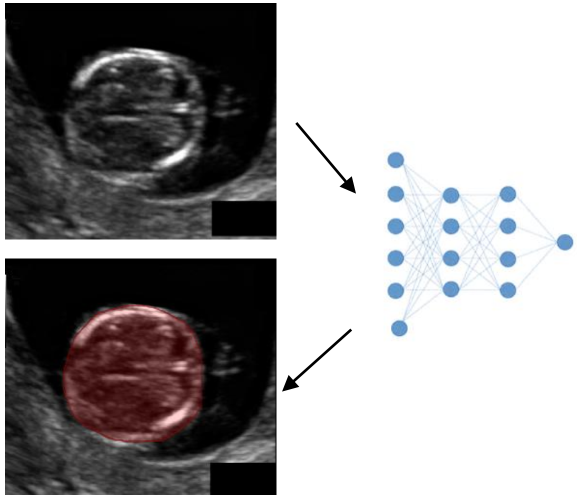
Model definition
Several deep learning networks

Nets
- *AHNet*
  - `AHNet`
- *DenseNet*
- *DenseNet121*
- *EfficientNet*
- *BlockArgs*
- *EfficientNetBN*
- *EfficientNetBNFeatures*
- *SegResNet*

Train model
Several losses and optimizers

Segmentation Losses
- *DiceLoss*
  - `DiceLoss`
  - `Dice`
  - `dice`
- *MaskedDiceLoss*
- *GeneralizedDiceLoss*
- *GeneralizedWassersteinDiceLos*
- *DiceCELoss*
- *DiceFocalLoss*

Test in new images



https://docs.monai.io/en/latest/